

$$g(x) = -4 \cos \left[\frac{\pi}{3} x + \frac{\pi}{6} \right] + 12$$

(6)

Application

1. A Ferris wheel rotates 1 time every 3 minutes. What is its angular velocity, in radians per second, in terms of π ?

$$(1) \quad \frac{1 \text{ rev}}{3 \text{ min}} \times \frac{\frac{2\pi \text{ rad}}{1 \text{ rev}}}{60 \text{ s}} = \frac{1}{90} \pi \text{ rad/s}$$

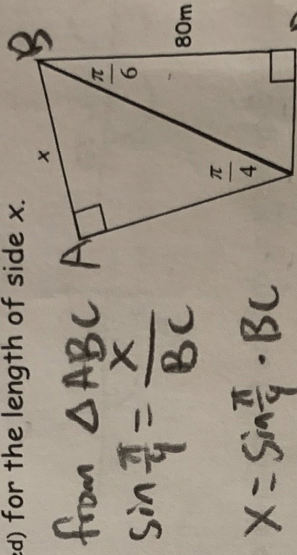
2. Simplify $\frac{\sin(x+y) + \sin(x-y)}{\cos(x+y) + \cos(x-y)}$.

$$(2) \quad \frac{(\sin x \cos y + \sin y \cos x) + (\sin x \cos y - \sin y \cos x)}{(\cos x \cos y - \sin x \sin y) + (\cos x \cos y + \sin x \sin y)} \\ = \frac{2 \sin x \cos y}{2 \cos x \cos y} = \frac{\sin x}{\cos x} = \tan x$$

3. Determine an exact expression (fully rationalized and reduced) for the length of side x .

$$\text{from } \triangle BCD \quad \frac{2}{\sqrt{3}} \cos \frac{\pi}{6} = \frac{80}{BC}$$

$$BC = \frac{80}{\cos \frac{\pi}{6}} = \frac{80}{\frac{\sqrt{3}}{2}} = \frac{160}{\sqrt{3}}$$



$$\text{from } \triangle ABC \quad \sin \frac{\pi}{4} = \frac{x}{BC}$$

$$x = \sin \frac{\pi}{4} \cdot BC$$

$$x = \frac{1}{\sqrt{2}} \cdot \left(\frac{160}{\sqrt{3}} \right)$$

$$x = \frac{160}{\sqrt{6}} = \frac{160\sqrt{6}}{6} = \frac{80\sqrt{6}}{3}$$

4. Prove the following identities

$$a) \frac{\sin 2x}{1 - \cos 2x} = \cot x$$

$$(1) \quad LS = \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)}$$

$$(2) \quad = \frac{2 \sin x \cos x}{2 \sin^2 x}$$

$$= \frac{\cos x}{\sin x}$$

$$= \cot x = RS$$

$$\therefore LS = RS$$

$$b) \csc \left(\frac{3\pi}{2} + x \right) = \frac{\cos x}{\sin^2 x - 1}$$

$$LS = \frac{1}{\sin \left(\frac{3\pi}{2} + x \right)}$$

$$= \frac{1}{\sin \frac{3\pi}{2} \cos x + \sin x \cos \frac{3\pi}{2}}$$

$$= \frac{1}{(-1) \cos x + 0} = \frac{\cos x}{-\cos^2 x}$$

$$= \frac{\cos x}{-(1 - \sin^2 x)} = \frac{\cos x}{\sin^2 x - 1} = RS$$

$$\therefore LS = RS$$

Communication

1. If $(\sin x)(\tan x)$ is negative, explain which quadrant(s) x can be in?

	$\sin x$	$\tan x$	$\sin x \tan x$
Q1	+	+	+
Q2	+	-	-
Q3	-	+	-
Q4	-	-	+

$\therefore$ it can be in Q2 or Q3

2. Explain why it is not possible to determine $\tan(x + \frac{\pi}{2})$ using a compound angle formula.

It's not possible because that would require evaluating

$\tan \frac{\pi}{2}$ which is $\frac{\sin \frac{\pi}{2}}{\cos \frac{\pi}{2}}$ and since $\cos \frac{\pi}{2} = 0$ it is

undefined (you can't divide by 0).

Thinking

1. If a and b are non-zero constants, fully simplify the following expression

$$\left(\sqrt{3}a \sin \frac{\pi}{b} \right)^2 + \left(\sqrt{3}a \cos \frac{\pi}{b} \right)^2$$

$$3a^2 \left(\sin^2 \frac{\pi}{b} + \cos^2 \frac{\pi}{b} \right)$$

$$= 3a^2 \left(\sin^2 \frac{\pi}{b} + \cos^2 \frac{\pi}{b} \right)$$

$$= 3a^2$$

2. Prove the following identities.

a) $\cot \theta - \tan \theta = 2 \cot 2\theta$

$$L.S. = \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{2 \cos 2\theta}{2 \sin \theta \cos \theta}$$

$$= \frac{2 \cos 2\theta}{\sin 2\theta}$$

$$= 2 \cot 2\theta$$

$$= R.S.$$

$$\therefore L.S. = R.S.$$

for one cycle on the grid provided.

KNOWLEDGE

b) $\frac{\cos 2x + 1}{\sin 2x} = \tan\left(\frac{\pi}{2} - x\right)$

(3) $LS = \frac{2\cos^2 x - 1 + 1}{2\sin x \cos x}$

$= \frac{\cos 2x}{\sin 2x}$

$= \cot x$

c) $\frac{\cos 2x}{1 + \sin 2x} = \frac{\cot x - 1}{\cot x + 1}$

$LS = \cos^2 x - \sin^2 x$

$\cos^2 x + \sin^2 x + 2\sin x \cos x$

$= (\cos x - \sin x)(\cos x + \sin x)$

$= \frac{(\cos x + \sin x)^2}{(\cos x - \sin x)}$

$= \frac{\cos x - \sin x}{\cos x + \sin x}$

$= \frac{\cos 2x + \sin 2x}{\sin 2x}$ if $\sin x - \cos x = 0.5$

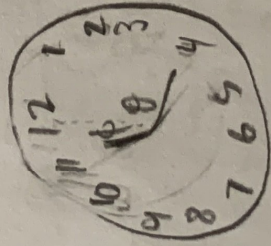
3. Determine $(\sin x - \cos x)^2 = \frac{1}{4}$

$\sin^2 x + \cos^2 x - 2\sin x \cos x = \frac{1}{4}$

$1 - \sin 2x = \frac{1}{4}$

$\sin 2x = \frac{3}{4}$

4. The hands of a clock show 11:20 am. Express the ~~acute~~ angle formed by the hour and minute hands in radian measure. Provide an exact solution. Smaller



ANS = $\theta + \phi = \frac{2\pi}{3} + \frac{\pi}{9} = \frac{7\pi}{9}$

angle min hand and $12(\theta)$ angle between

$\theta = 12 \text{ min} \times \frac{\text{rev}}{60 \text{ min}} \times \frac{2\pi}{\text{rev}}$

$= \frac{2\pi}{3}$

$\phi = \left(11 - 12 \text{ min} \times \frac{h}{60 \text{ min}}\right) \times \frac{\text{rev}}{12h} \times \frac{2\pi}{\text{rev}}$

$= \frac{2\pi}{9}$

$RS = \frac{\sin(\frac{\pi}{2} - x)}{\cos(\frac{\pi}{2} - x)}$

$= \frac{\sin \frac{\pi}{2} \cos x - \sin x \cos \frac{\pi}{2}}{\cos \frac{\pi}{2} \cos x + \sin x \sin \frac{\pi}{2}}$

$= \frac{\cos x - 0}{0 + \sin x}$

$\therefore LS = RS$

$= \cot x$

$\rightarrow = \frac{(\cos x - \sin x)}{(\cos x + \sin x)} \times \frac{1}{\sin x}$

$= \frac{\cos x - 1}{\sin x}$

$= \frac{\cos x}{\sin x} + 1$

$= \frac{\cot x - 1}{\cot x + 1} = RS$

$\therefore LS = RS$

Total
35 / 35

DRAW ALL YOUR DIAGRAMS FOR FULL MARKS!

Knowledge

1. Use the compound angle identity formulas to evaluate, exact value, for the following:

(Do not rationalize)

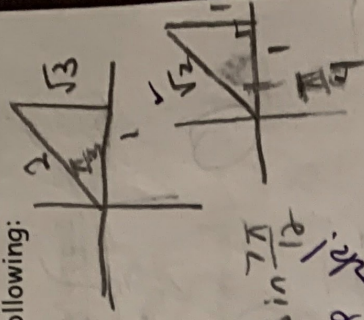
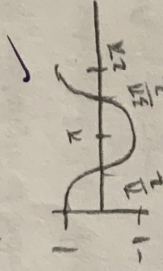
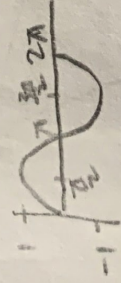
a) $\sin \frac{2\pi}{9} \cos \frac{5\pi}{18} + \cos \frac{2\pi}{9} \sin \frac{5\pi}{18}$

b) $\cos \frac{13\pi}{12}$

(1) $\sin \left(\frac{2\pi}{9} + \frac{5\pi}{18} \right)$

$= \sin \frac{\pi}{2}$

$= 1$



$\cos \left(\frac{\pi}{2} + \frac{7\pi}{12} \right)$

$= \cos \frac{\pi}{2} \cos \frac{7\pi}{12} - \sin \frac{\pi}{2} \sin \frac{7\pi}{12}$

$= 0 - (1) \sin \frac{7\pi}{12}$

$= -\sin \left(\frac{\pi}{3} + \frac{\pi}{4} \right)$

$= -\left(\sin \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{4} \cos \frac{\pi}{3} \right)$

$= -\left(\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \right) = -\frac{1}{2\sqrt{2}} \left(\sqrt{3} + 1 \right)$

$= -\frac{\sqrt{6} + \sqrt{2}}{4}$

2. Express each of the following as a single trigonometric function, simplify fully.

a) $8 \sin 3x \cos 3x$

(2) $4(2 \sin 3x \cos 3x)$

$= 4 \sin 6x$

$4(1 - 2 \sin^2 \left(\frac{\pi}{2} \right))$

$= 4 \cos \alpha$

3. Angle
- x
- is in the second quadrant such that
- $\tan x = -\frac{12}{5}$

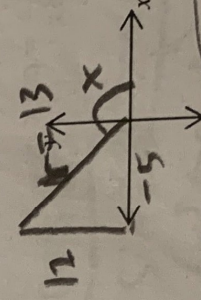
- a) Draw a sketch that represents angle
- x
- .

- b) Determine the
- exact value
- for
- $\frac{\sin x}{\cos 2x}$
- .

$\frac{\sin x}{\cos 2x} = \frac{\sin x}{1 - 2 \sin^2 x}$

$= \frac{\left(\frac{12}{13} \right)}{1 - 2 \left(\frac{12}{13} \right)^2}$

$= \frac{\frac{12}{13}}{\frac{169 - 288}{169}}$



$\sin x = \frac{12}{r} = \frac{12}{13}$

$r^2 = 12^2 + (-5)^2 = 169$
 $r = 13$

$= \frac{-156}{119}$